Arrival and required times

Once the graph is levelized, you propagate tags

Goldens to remember

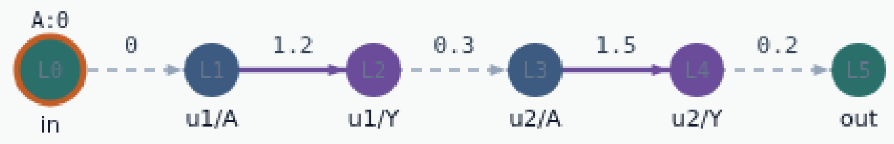
- Forward: A(in)=0, A(u1/Y)=1.2, A(out)=3.2
- Backward setup: R(out)=10, R(in)=6.8
- Keep these numbers handy, the browser challenges and Track A tests use the same instance
- <!-- algorithm-walkthrough -->

Seed arrival at the source

ARRIVAL AND REQUIRED TIMES

Seed arrival at the source

Step 1 / 5 · seed-arrival · module01-03-arrival-required



Explanation

Set arrival at in to zero for the launch edge. Every other pin waits for its predecessors.

- Arrival at sources starts the wavefront
- Here A(in) = 0
- No delay yet on the first net

```
A(in) = 0
```

Seed arrival at the source

Propagate arrival forward

ARRIVAL AND REQUIRED TIMES

Propagate arrival forward

Step 2 / 5 · forward-wave · module01-03-arrival-required

Explanation

For each pin in topo order, arrival is the max over predecessors of A(pred) + delay. At out the wavefront reaches 3.2.

- A(v) = max(A(u) + d(u → v))
- Cell delays dominate this toy path
- Golden A(out) = 3.2

```
A(u1/Y)=1.2
A(u2/A)=1.5
A(u2/Y)=3.0
A(out)=3.2
```

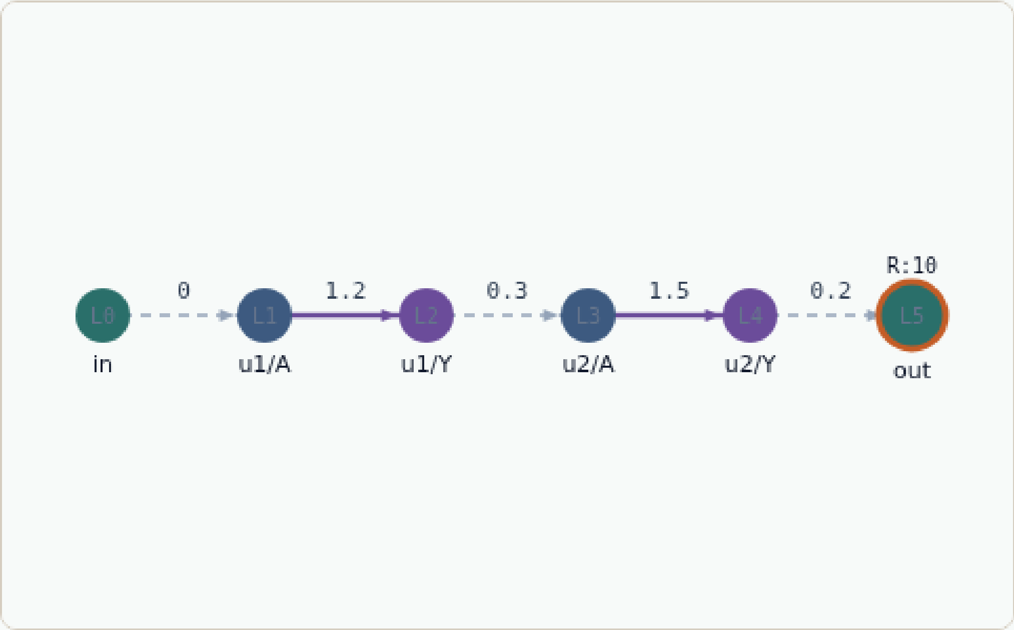
Propagate arrival forward

Seed required at the sink

ARRIVAL AND REQUIRED TIMES

Seed required at the sink

Step 3 / 5 · seed-required · module01-03-arrival-required



Explanation

For a single-cycle setup check, required at out equals the clock period—here 10.

- Setup capture edge sets R(out)
- R(out) = period × cycles
- Default cycles = 1 → 10

```
period = 10
R(out) = 10
```

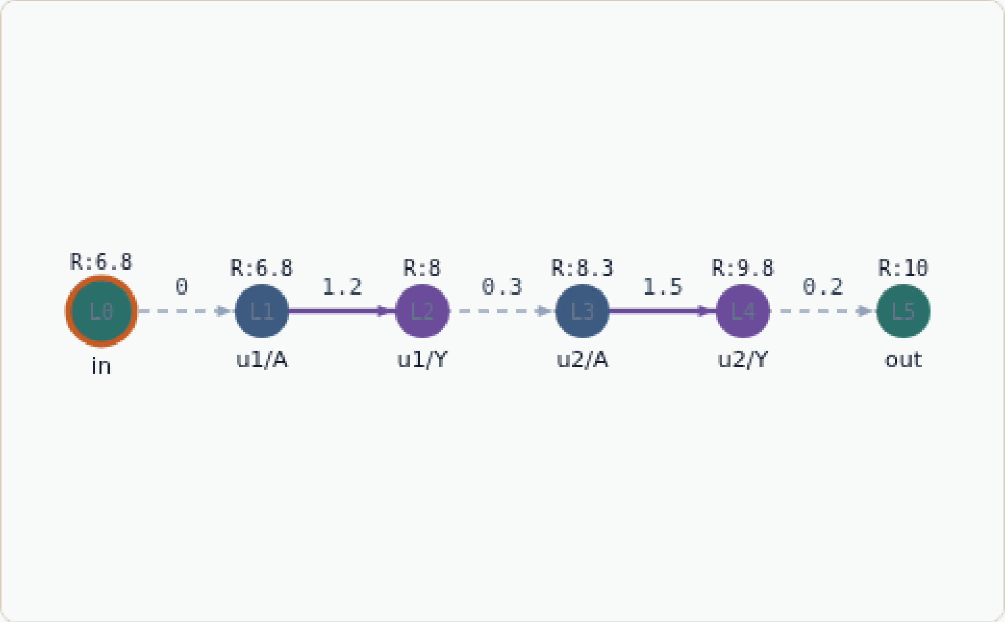
Seed required at the sink

Propagate required backward

ARRIVAL AND REQUIRED TIMES

Propagate required backward

Step 4 / 5 · backward-wave · module01-03-arrival-required



Explanation

Walk reverse topo order. Required at a pin is the min over successors of R(succ) − delay. At in, required becomes 6.8.

- $R(u) = \min(R(v) - d(u \rightarrow v))$
- Tightest successor wins
- Golden R(in) = 6.8

```
R(u2/Y)=9.8
R(u1/Y)=8.0
R(u1/A)=6.8
R(in)=6.8
```

Propagate required backward

Keep both tags on every pin

ARRIVAL AND REQUIRED TIMES

Keep both tags on every pin

Step 5 / 5 · both-tags · module01-03-arrival-required

Explanation

Arrival and required live together. Slack at a pin is R − A for setup. Next labs turn those tags into slack and a critical path.

- Forward then backward
- Same graph, two tag maps
- A(out)=3.2 and R(out)=10

```
A(out)=3.2
R(out)=10
next: slack = 6.8
```

Keep both tags on every pin

Browser lab track

- In the browser lab, open **arrival-required**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

Implement track

- In the implement track
- Run `python3 common/test_propagate.py` (and the timing-graph test) until the goldens print

Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue